CHAPTER 1

Functions and Their Graphs

Functions are a major part of science, engineering, and math. You can think of a function as a machine: you put something into the machine, it processes it, and out comes something else: a product. Just as we often use the variable x to stand in for a number, we often use the variable f to stand in for a function. $f(x)$ is said as "f of x". However, this could be any letter or word that takes an input x, maybe even $\text{zebra}(x)$!

For example, we might ask, "Let the function f be defined like this:

$$f(x) = -5x^2 + 12x + 2$$

What is the value of $f(3)$, said as "f of 3"?

You would run the number 3 through "the machine": $-5(3^2)+12(3)+2 = -7$. The answer would be "$f(3)$ is 7".

However, some functions are not defined for every possible input. For example:

$$f(x) = \frac{1}{x}$$

This is defined for any x except 0, because you can't divide 1 by 0. The set of values that a function can process is called its *domain*, and resulting output values are called the range. It is important to note that functions have only one *output* for each *input*. However, multiple inputs can have the same output. A relationship where one input can result in the more than one output is not a function, but a relation. This can be proven by the vertical line test.

Exercise 1 Domain of a function

Working Space

Let the function f be given by $f(x) = \sqrt{x-3}$. What is its domain?

Answer on Page 25

1.1 Graphs of Functions

If you have a function, f, its graph is the set of pairs (x, y) such that $y = f(x)$. We usually draw a picture of this set, called a *graph*. The graph not only includes the picture, but also the values of x and y used to create it.

Here is the graph of the function $f(x) = -5x^2 + 12x + 2$:

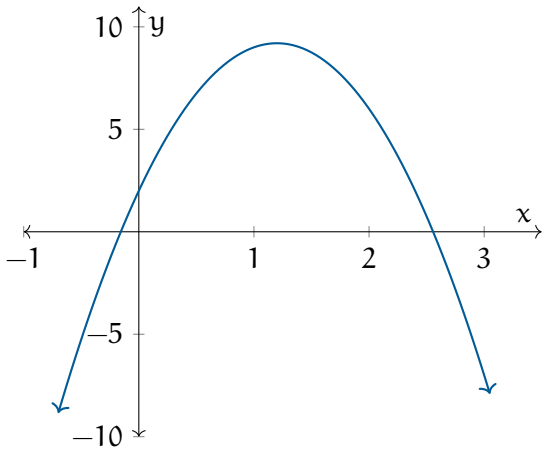


Figure 1.1: The function $f(x) = -5x^2 + 12x + 2$ *graphed* on an xy-plane from $x = -1$ to $x = 3$. The inputs x are graphed along the x-axis, and the output of the function $f(x)$ is plotted along the y-axis.

(Note that this is just part of the graph; it goes infinitely in both directions. Remember your vectors!)

Here is the graph of the function $f(x) = \frac{1}{x}$:

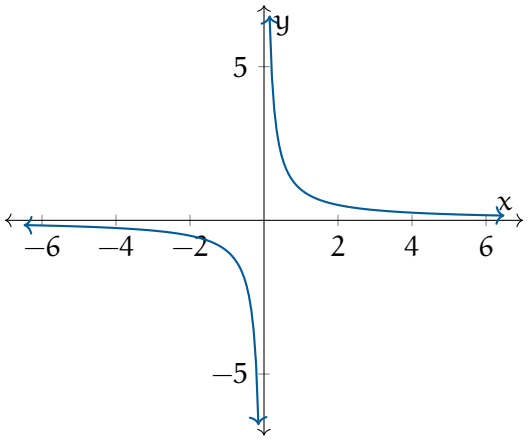


Figure 1.2: The function $f(x) = \frac{1}{x}$ graphed on a coordinate plane.

Notice the *asymptotes* at $y = 0$ and $x = 0$. Here, the function is undefined at $x = 0$ and $y = 0$, but is continually approaching it. It never actually reaches these values though! $f(0)$ is not defined as you cannot divide by zero (recall division rules) and as x is increasing, $f(x)$ gets infinitesimally small but never equal to zero. We will talk more about these in the rational functions chapter.

To draw a graph, take each x value (usually in increments of 1) and determine its corresponding y value. Then, plot those points on a graph, using the origin as the starting point.

Exercise 2 Draw a graph

Working Space

Let the function f be given by $f(x) = -3x + 3$. Sketch its graph.

Answer on Page 25

1.2 Can this be expressed as a function?

Note that not all sets can be expressed as graphs of functions. For example, here is the set of points (x, y) such that $x^2 + y^2 = 9$:

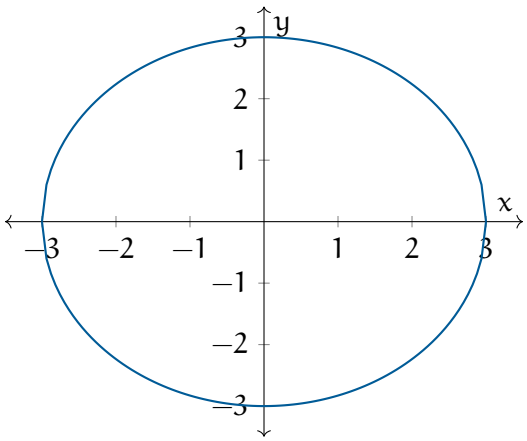


Figure 1.4: The graph of $x^2 + y^2 = 9$.

This cannot be the graph of a function, because what would $f(0)$ be? 3 or -3? This set fails what we call "the vertical line test": If any vertical line contains more than one point from the set, it isn't the graph of a function. For example, the vertical line $x = 2$ would cross the graph twice:

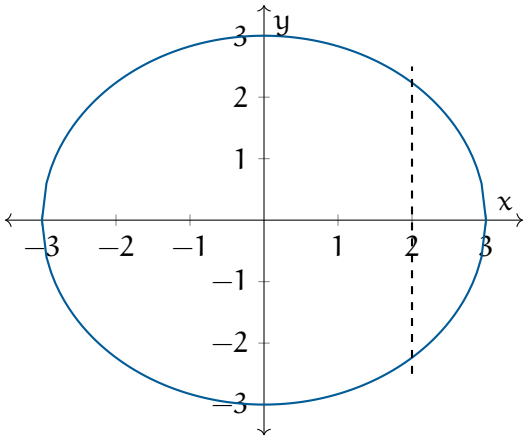


Figure 1.5: Vertical line test on $x^2 + y^2 = 9$.

1.3 Inverses

Some functions have inverse functions. If a function f is a machine that turns number x into y, the inverse (usually denoted f^{-1}) is the machine that turns y back into x.

For example, let $f(x) = 5x + 1$. Its inverse is $f^{-1}(x) = (x - 1)/5$. (Spot check it: $f(3) = 16$ and $f^{-1}(16) = 3$)

Does the function $f(x) = x^3$ have an inverse? Yes, $f^{-1}(x) = \sqrt[3]{x}$. Let's plot the function (solid line) and its inverse (dashed):

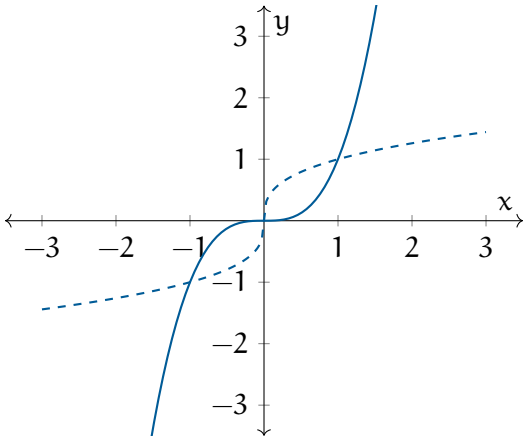


Figure 1.6: The function $f(x) = x^3$ and it's inverse.

The inverse is the same as the function, just with its axes swapped. The inverse function is *reflected* across the line $y = x$. This tells us how to solve for an inverse: We swap x and y and solve for y.

For example, if you are given the function $f(x) = 5x + 1$, its graph is all (x, y) such that $y = 5x + 1$. The graph of its inverse is all (x, y), such that $x = 5y + 1$. Solving for y gives you $y = (x - 1)/5$. So we can say that $f^{-1}(x) = (x - 1)/5$, QED.

Not every function has an inverse. For example, $f(x) = x^2$. Note that $f(2) = f(-2) = 4$. What would $f^{-1}(4)$ be? 2 or -2? This implies the "horizontal line test": If any horizontal line contains more than one point of a function's graph, that function has no inverse. If a function passes the horizontal line test, it is called "one-to-one", meaning there is exactly one x that gives each y.

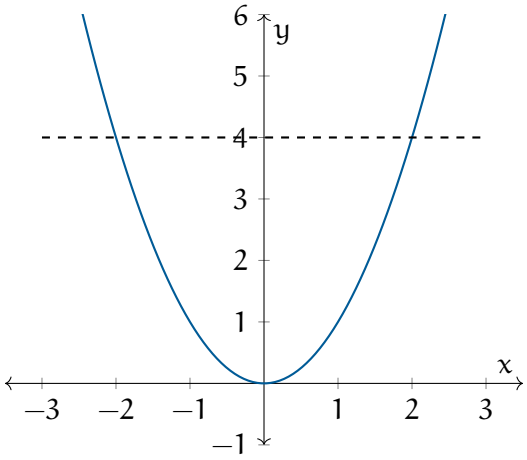


Figure 1.7: The horizontal line test run on $f(x) = x^2$.

In some problems, you need an inverse, but you don't need the whole domain, so you trim the domain to a set you can define an inverse on. This allow you to make claims such as "If we restrict the domain to the nonnegative numbers, the function $f(x) = x^2 - 5$ has an inverse: $f^{-1}(x) = \sqrt{x + 5}$.

This raises the question: What is the domain of the inverse function f^{-1}?

If we let X be the domain of f, we can run every member of X through "the machine" and gather them in a set on the other side. This set would be the *image* of the f "machine". (This is the *range* of f.)

What is the image of $f(x) = x^2 - 5$? It is the set of all real numbers greater than or equal to -5. We write this:

$$\{x \in \mathbb{R} | x \geq -5\}$$

Now we can say: **The range (or image) of the function is the domain of the inverse function.**

In inverse functions, the domain and range get swapped: the domain of the function is the range of the inverse function, and visa versa. In our example, we can use any number greater than or equal to -5 as input into the inverse function.

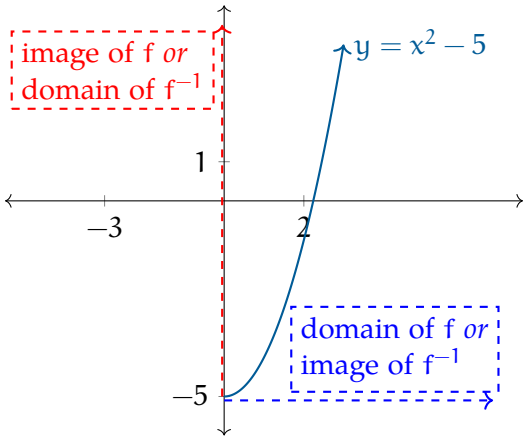


Figure 1.8: Domain and range of functions and their inverses get swapped.

Exercise 3 Find the inverse

Working Space

Let $f(x) = (x-3)^2 + 2$. Sketch the graph.

Using all the real numbers as a domain, does this function have an inverse?

How would you restrict the domain to make the function invertible?

What is the inverse of that restricted function?

What is the domain of the inverse?

Answer on Page 25

Exercise 4

A function is given by a table of values, a graph, or a written description. Determine whether it is one-to-one.

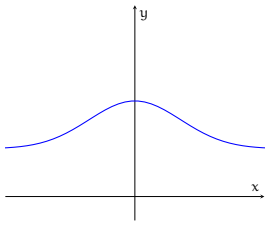
Working Space

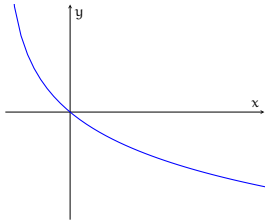
1.

x	1	2	3	4	5	6
f(x)	1.5	2.0	3.6	5.3	2.8	2.0

2.

x	1	2	3	4	5	6
f(x)	1.0	1.9	2.8	3.5	3.1	2.9

3. 

4. 

5. $f(t)$ is the height of a football t seconds after kickoff

6. $v(t)$ is the velocity of a dropped object

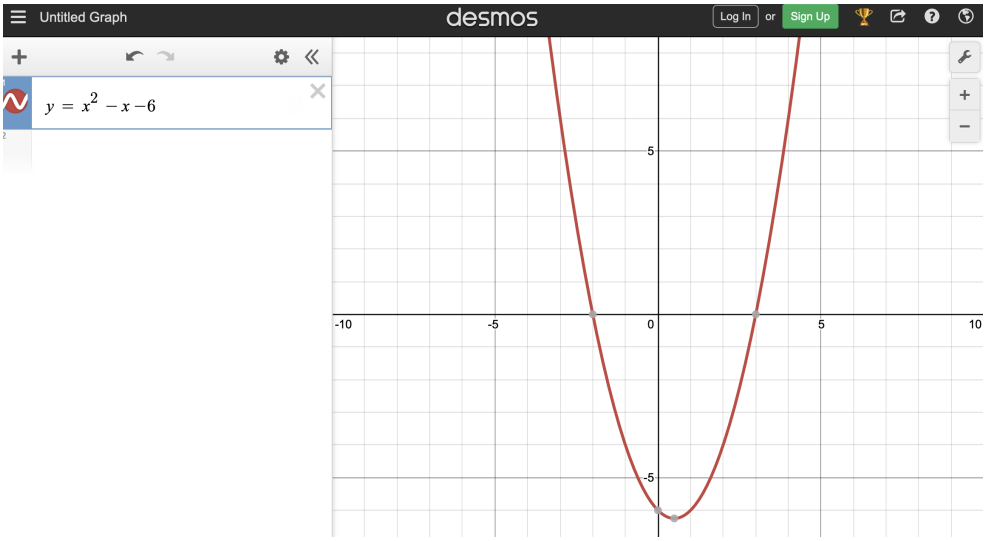
Answer on Page 26

1.4 Graphing Calculators

One really easy way to understand your function better is to use a graphing calculator. Desmos is a great, free online graphing calculator.

In a web browser, go to Desmos: `https://www.desmos.com/calculator`

In the field on the left, enter the function $y = x^2 - x - 6$. (For the exponent, just prefix it with a caret symbol: "x^2".)



Computing in Python

Python can draw the same graph. The listing below uses two libraries: NumPy, which builds lists of numbers called *arrays*, and Matplotlib, which plots them.

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return x**2 - x - 6

x = np.linspace(-5, 6, 200)
y = f(x)

plt.plot(x, y)
plt.axhline(0, color="black", linewidth=0.8)
plt.axvline(0, color="black", linewidth=0.8)
plt.grid(True)
plt.xlabel("x")
plt.ylabel("y")
plt.title("y = x^2 - x - 6")
plt.show()
```

The `def` line packages the rule $x^2 - x - 6$ into a Python function named `f`. Calling `f(3)` runs 3 through the machine, the same way $f(3)$ does on paper. The call `np.linspace(-5, 6, 200)` creates 200 evenly spaced x values from -5 to 6, and passing that whole array to `f` returns the 200 matching y values at once. Matplotlib then draws a point at each (x, y) pair and joins neighboring points with short line segments. With 200 points, each

segment is short enough that the curve looks smooth. The calls `axhline` and `axvline` draw the x-axis and y-axis, and `plt.show()` opens the plot window.

Exercise 5 Reading values from f

Add this line after the plot code:

Working Space

```
print(f(0), f(3), f(-2))
```

What does it print? What do the three outputs tell you about the graph?

Answer on Page 27

Desmos handles the asymptote of $\frac{1}{x}$ on its own. Matplotlib joins every pair of neighboring points, so the two branches of $\frac{1}{x}$ need separate arrays, one on each side of $x = 0$:

```
x_left = np.linspace(-6, -0.15, 100)
x_right = np.linspace(0.15, 6, 100)

plt.plot(x_left, 1 / x_left, color="C0")
plt.plot(x_right, 1 / x_right, color="C0")
plt.axhline(0, color="black", linewidth=0.8)
plt.axvline(0, color="black", linewidth=0.8)
plt.ylim(-7, 7)
plt.grid(True)
plt.title("y = 1/x")
plt.show()
```

Both calls use `color="C0"`, Matplotlib's default blue, so the two branches read as one function. The call `plt.ylim(-7, 7)` fixes the visible y range to match Figure 1.2.

Exercise 6 The phantom line

A student plots $\frac{1}{x}$ with a single array:

```
x = np.linspace(-6, 6, 100)
plt.plot(x, 1 / x)
plt.show()
```

The graph shows a vertical line running up the y-axis. The function has no value at $x = 0$. Where does the line come from?

Working Space

Answer on Page 27

The inverse figure from earlier in this chapter takes three plot calls: the function, its inverse, and the line $y = x$ that reflects one onto the other.

```
def g(x):
    return x**3

def g_inv(x):
    return np.cbrt(x)

x = np.linspace(-3, 3, 200)

plt.plot(x, g(x), label="g(x) = x^3")
plt.plot(x, g_inv(x), linestyle="--", label="inverse")
plt.plot(x, x, color="gray", linewidth=0.8, label="y = x")
plt.xlim(-3.5, 3.5)
plt.ylim(-3.5, 3.5)
plt.gca().set_aspect("equal")
plt.grid(True)
plt.legend()
plt.show()
```

The function `np.cbrt` returns the cube root of every entry in an array. Setting the aspect to `"equal"` gives one unit the same length on both axes. Without it, the window stretches one axis and the reflection across $y = x$ looks lopsided. The `label` strings feed `plt.legend()`, which prints the key in the corner.

Exercise 7 The missing half

Replace `np.cbrt(x)` with `x**(1/3)` and run the code again. The dashed curve now appears only to the right of the y-axis, and Python prints the warning `invalid value encountered in power`. Explain what happened.

Working Space

Answer on Page 27

1.5 Even and Odd Functions

1.5.1 Even Functions

An even function is symmetric about the y-axis. That means if you fold the graph along the y-axis, both sides will match perfectly. Note that the input value yields the same output regardless of whether the input value is positive or negative.

Even Functions

A function $f(x)$ is even if

$$f(-x) = f(x)$$

for all x in its domain.

Examples:

- $f(x) = x^2$
- $f(x) = \cos(x)$
- any $f(x) = x^n$ where n is an even number
- $f(x) = |x|$

On a graph, a function will be a mirror image across the y-axis.

1.5.2 Odd Functions

An odd function is symmetric about the origin. That means if you rotate the graph 180° about the origin, it lands on itself. Algebraically, when you input a value k, you get some output n; if you negate that input as $-k$, the output is also negated resulting in $-n$.

Odd Functions

A function $f(x)$ is odd if

$$f(-x) = -f(x)$$

for all x in its domain.

Equivalently,

$$f(x) + f(-x) = 0$$

for all x in its domain.

- $f(x) = x^3$
- $f(x) = \sin(x)$
- $f(x) = \tan(x)$
- any $f(x) = x^n$ where n is an odd number

On a graph, rotating the graph 180° about the origin will yield the same graph.

1.5.3 Neither Even Nor Odd

Some functions are neither even nor odd.

For example, $f(x) = x^3 + 1$ does not satisfy either condition. **Computing in Python**

A plot can test for symmetry. Draw $h(x)$, $h(-x)$, and $-h(-x)$ on one set of axes. When the $h(-x)$ curve lies on top of $h(x)$, the function is even. A match between $-h(-x)$ and $h(x)$ means the function is odd.

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def h(x):
5      return x**3 + 1
6
```

```
x = np.linspace(-2, 2, 200)

plt.plot(x, h(x), label="h(x)")
plt.plot(x, h(-x), linestyle="--", label="h(-x)")
plt.plot(x, -h(-x), linestyle=":", label="-h(-x)")
plt.grid(True)
plt.legend()
plt.show()
```

Writing `h(-x)` negates every entry of the array before it reaches `h`, and the extra minus sign in `-h(-x)` negates every output afterward. For $h(x) = x^3 + 1$, the dashed and dotted curves both separate from the solid one, so $x^3 + 1$ is neither even nor odd.

The function `np.allclose(a, b)` turns the visual check into a yes or no answer. It returns `True` when two arrays agree at every entry, allowing for tiny rounding errors:

```
print(np.allclose(h(-x), h(x)))   # even test
print(np.allclose(-h(-x), h(x)))  # odd test
```

Both lines print `False` for $h(x) = x^3 + 1$.

Exercise 8 One input is enough

Predict the output, then run it:

```
print(h(1), h(-1), -h(-1))
```

How do these three numbers show that `h` is neither even nor odd?

Working Space

Answer on Page 27

Exercise 9 Swap the function

Change the body of `h` to `return np.abs(x)` and run the plot and both `allclose` tests. Then change it to `return x**3` and run everything again. For each version, which curve lands on top of the solid one, and what do the tests print?

Working Space

Answer on Page 27

1.6 Piecewise Functions

Another type of function that will be integral to your mathematics career are *piecewise functions*. A piecewise function is one defined by two different equations. You chose which function to evaluate dependent on the domain. Let's look at a simple example:

$$f(x) = \begin{cases} \frac{1}{x} & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

This piecewise function has two cases: when $x < 0$, we use the equation $\frac{1}{x}$; when $x \geq 0$, we use x^2. Notice that we don't consider $x = 0$ for $\frac{1}{x}$ since it would be undefined, but it is defined under x^2. The domain of $f(x)$ is defined for all values of x! Let's make a table of values:

x	-2	-1	0	0.1	1	2
$f(x)$	$-\frac{1}{2}$	-1	0	0.01	1	4

We can graph this like the following

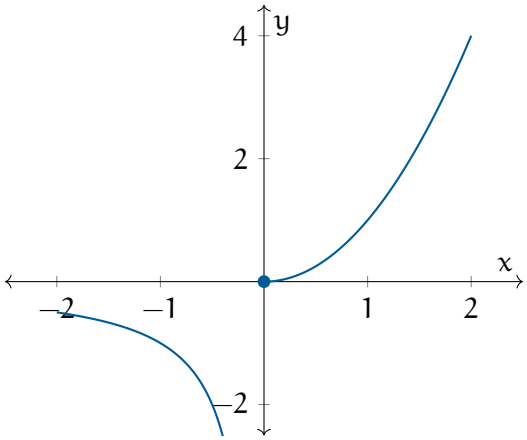


Figure 1.10: The piecewise function $f(x)$ graphed.

Notice the *closed circle*. This indicates that the value at the is included at a disjunction. An *open circle* would indicate that the value at that point is not included. Say we have the following function:

$$g(x) = \begin{cases} x^2 + 1 & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

Graphing this, we obtain an open circle at $(0, 1)$ and a closed circle at $(0, 0)$.

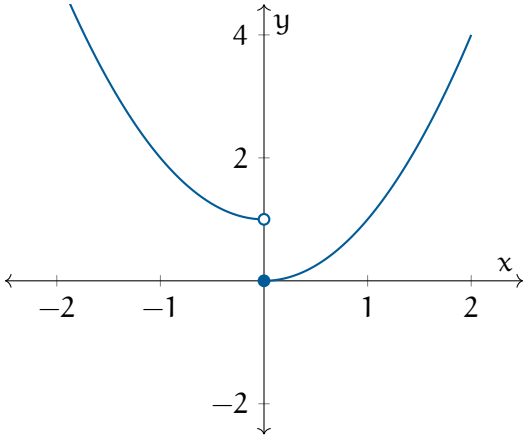


Figure 1.11: The piecewise function $g(x)$ graphed.

Exercise 10 Absolute Value

Working Space

Rewrite the absolute value function $|x|$ as a piecewise function.

Answer on Page 28

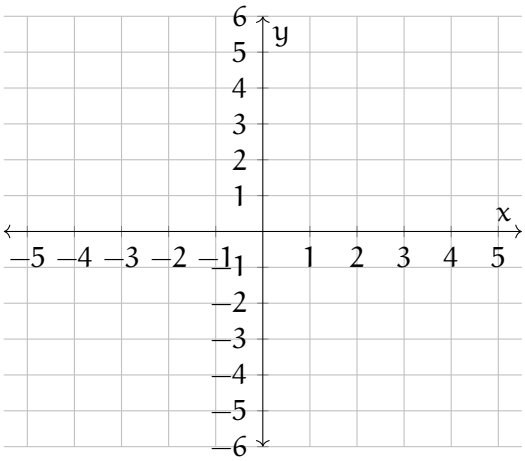
Exercise 11 Piecewise 1

Working Space

$$f(x) = \begin{cases} x+2 & \text{if } -5 \le x < -1 \\ x^2 & \text{if } -1 \le x \le 2 \\ 3-x & \text{if } 2 < x \le 5 \end{cases}$$

For the equation above, create a table for every integer from -5 to 5 (inclusive), and graph.

Here is a blank coordinate plane for plotting the piecewise function:



Answer on Page 28

1.6.1 Floor and Ceiling Functions

Another common piecewise function you may see are the *floor function* and *ceiling function*. The floor function, which takes the any real number x, is defined as

$$\lfloor x \rfloor \text{ or } \text{floor}(x) = \begin{cases} n & \text{if } n \le x < n+1, \quad n \in \mathbb{Z} \end{cases}$$

This means that you take any number x and find the closest integer less than or equal to it. For example, $\lfloor 0.7 \rfloor$ would be floored to 0.

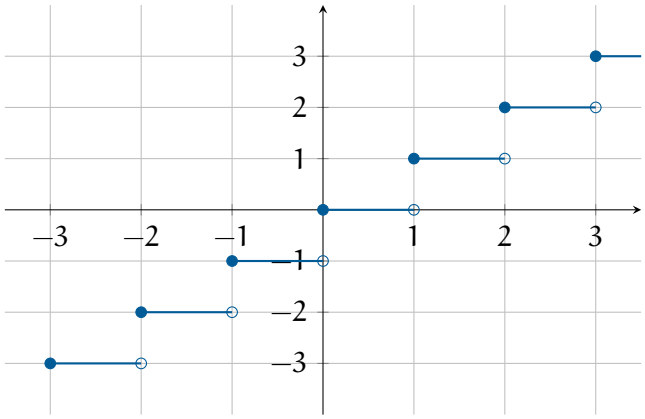


Figure 1.12: The flooring function, $f(x) = \lfloor x \rfloor$sdkblue,

Similarly, the ceiling function is defined as follows:

$$\lceil x \rceil \text{ or } \operatorname{ceil}(x) = \begin{cases} n & \text{if } n-1 < x \leq n, \quad n \in \mathbb{Z} \end{cases}$$

The ceiling function takes in any real numbers and returns the next greatest integer. For example, $\lceil 8.00001 \rceil$ returns 9, the next greatest integer. For any valid integer like $x = 7$, the $\lfloor 7 \rfloor$ and $\lceil 7 \rceil$ are both equal to 7.

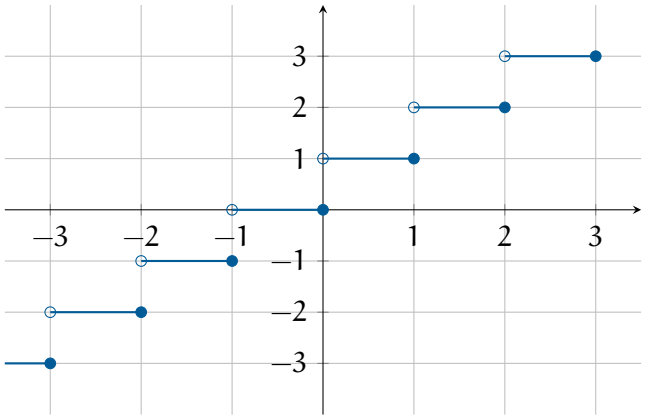


Figure 1.13: The ceiling function, $f(x) = \lceil x \rceil$.

Exercise 12 More TVs is Better

Working Space

A store sells televisions with pricing that depends on how many units a customer buys:

- For the first 5 TVs, the cost per TV is \$500
- For the next 10 TVs, the cost per TV is \$350
- Any additional TVs cost \$300

A hotel is buying TVs for each hotel room.

1. Construct a piecewise function $C(x)$ for the cost of TVs. Assume TVs must be a discrete integer (you cannot buy 8.5 TVs).
2. Find the cost of 12 TVs using your function.
3. When is average cost $<$ \$360?

Answer on Page 28

1.6.2 Invalid Piecewise Functions

One thing to note when creating piecewise functions is to make sure your cases do not contradict each other, or in other words, create cases that define two different possibilities for x at a given value. For example,

$$f(x) = \begin{cases} 10.2 & x = 4 \\ x^2 & x \geq 0 \end{cases}$$

This creates an impossible and thus invalid piecewise function, as at $x = 4$, $f(x)$ is both defined as 10.2 and 16. This would be impossible ($\rightarrow\leftarrow$). In this case, either you define

the value at DNE or undefined, or revise the piecewise function at that point. Most often, it is done as follows:

$$f(x) = \begin{cases} 10.2 & x = 4 \\ x^2 & x \geq 0,\ x \neq 4 \end{cases}$$

Then our value at $x = 4$ is equal to 10.2.

Computing in Python

Matplotlib draws a piecewise graph one piece at a time. Here is $g(x)$ from earlier in this section, with its open and closed circles:

```
import math
import numpy as np
import matplotlib.pyplot as plt

x_left = np.linspace(-2, 0, 100)
x_right = np.linspace(0, 2, 100)

plt.plot(x_left, x_left**2 + 1, color="C0")
plt.plot(x_right, x_right**2, color="C0")
plt.plot(0, 0, "o", color="C0")                              # closed circle
plt.plot(0, 1, "o", color="C0", markerfacecolor="white")  # open circle
plt.grid(True)
plt.title("piecewise g(x)")
plt.show()
```

Each piece gets its own array covering only its part of the domain. A single array across the whole domain would join the two pieces with a vertical segment at $x = 0$, the same phantom line that appears with $\frac{1}{x}$. The format string `"o"` draws one round marker, and `markerfacecolor="white"` hollows it out into an open circle.

Plotting shows the shape. Evaluating the function at a particular input takes `if` and `elif`, with one condition for each row of the cases. This is the function from Exercise 11:

```
def p(x):
    if -5 <= x < -1:
        return x + 2
    elif -1 <= x <= 2:
        return x**2
    elif 2 < x <= 5:
        return 3 - x
    return None

for x in range(-5, 6):
    print(x, p(x))
```

Python accepts chained comparisons such as `-5 <= x < -1`, written exactly in math notation. It checks the conditions from top to bottom and returns from the first one that is true. An input outside every case, such as `p(6)`, reaches the last line and returns `None`, Python's value for "nothing." That matches the function being undefined there. The call `range(-5, 6)` produces the integers from -5 up to 5; the stop value 6 is left out.

Exercise 13 Checking the table

Run the loop above. Compare its output with the table in the answer to Exercise 11. Which case does Python use at $x = -1$ and at $x = 2$?

Working Space

Answer on Page 29

Exercise 14 The case that never runs

A student codes the revised function from the end of this section:

Working Space

```
def q(x):
    if x >= 0:
        return x**2
    elif x == 4:
        return 10.2
```

The call `q(4)` returns `16` instead of `10.2`. Find the error and fix it.

Answer on Page 29

Exercise 15 Floor and ceiling in Python

Python's `math` module provides `math.floor` and `math.ceil`. Predict the output:

```
print(math.floor(-2.5),
↪  math.ceil(-2.5),
↪  math.ceil(8.00001),
↪  math.floor(7))
```

Working Space

Answer on Page 30

1.7 Summary

This chapter talked about functions. Functions take in an input and spit out an output, like a magician putting a banana in a hat and releasing a white bunny. However, this is not magic, but math. Each input is given exactly one output. Not all functions have equations that are defined for all real numbers.

We can also graph functions according to their inputs (the x values on the x-axis) and outputs (the returns of the function $f(x)$ plotted on the y axis).

We established that relations must pass the vertical line test to be functions. Some functions have inverse functions, which reverse the action of the original function. A function has an inverse only if it is one-to-one, which can be checked with the horizontal line test. When a function and its inverse are graphed, they are reflections of each other across the line $y = x$ and the domain and range of a function switch roles in its inverse.

Functions can also be classified by symmetry (even or odd). Piecewise functions are by different formulas on different parts of the domain.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

You can only take the square root of nonnegative numbers, so the function is only defined when $x - 3 \geq 0$. Thus, the domain is all real numbers greater than or equal to 3.

Answer to Exercise 2 (on page 4)

The graph of this function is a line, its slope is -3, and it intersects the y axis at $(0, 3)$.

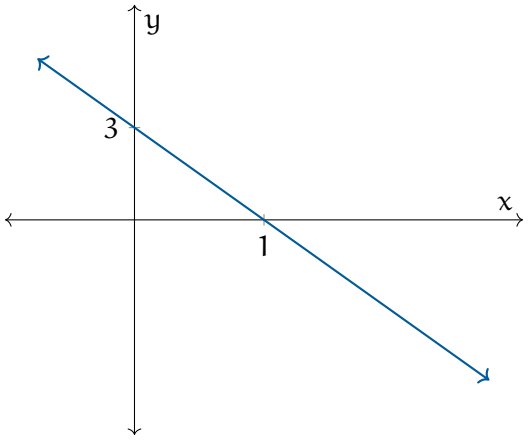


Figure 1.3: The resulting graph of $f(x) - 3x + 3$.

Answer to Exercise 3 (on page 7)

This graph is the graph of $y = x^2$ that has been moved to the right by three units and up two units:

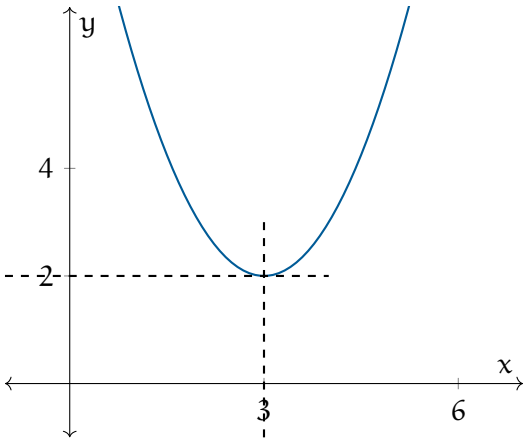


Figure 1.9: The graph of $f(x) = (x-3)^2 + 2$.

To prevent any horizontal line from containing more than one point of the graph, you would need to use the left or the right side — either $\{x \in \mathbb{R} | x \leq 3\}$ or $\{x\mathbb{R} | x \geq 3\}$. Most people will choose the right side; the rest of the solution will assume that you did too.

To find the inverse we swap x and y: $x = (y-3)^2 + 2$

Next, we solve for y to get the inverse: $y = \sqrt{x-2} + 3$

You can take the square root of nonnegative numbers. So the function $f^{-1}(x) = \sqrt{x-2}+3$ is defined whenever x is greater than or equal to 2.

Answer to Exercise 4 (on page 8)

1. This function is not one-to-one. From $x = 3$ to $x = 4$, the function increases from 3.6 to 5.3, which means it must pass through $f(x_1) = 4.0$. From $x = 4$ to $x = 5$, the function decreases from 5.3 to 2.8, which means it must pass through $f(x_2) = 4.0$ again.
2. This function is not one-to-one by a similar argument in the above solution
3. This function is not one-to-one, because it fails the horizontal line test
4. This function is one-to-one, because it passes the horizontal line test
5. $f(t)$ would not be one-to-one because the football must pass through each height (except the peak height) both on the way up and on the way back down
6. $v(t)$ would be one-to-one because a falling object only speeds up. Therefore, every time has a unique speed.

Answer to Exercise 5 (on page 10)

It prints `-6 0 0`. The first value says the graph crosses the y-axis at $(0, -6)$. The two zeros say $x = 3$ and $x = -2$ are the x-intercepts, the points where the parabola crosses the x-axis. Both match the Desmos graph.

Answer to Exercise 6 (on page 11)

This array skips 0. The two entries closest to it are $x \approx -0.0606$ and $x \approx 0.0606$, with $y \approx -16.5$ and $y \approx 16.5$. Matplotlib connects neighbors, so it draws a nearly vertical segment from $(-0.0606, -16.5)$ up to $(0.0606, 16.5)$. That segment belongs to neither branch of the graph. Plotting each side of the asymptote separately removes it. (With 101 points instead of 100, the array contains 0 exactly; NumPy then warns `divide by zero` and stores `inf` for that entry.)

Answer to Exercise 7 (on page 12)

NumPy refuses to raise a negative decimal number to a fractional power, so every negative entry of `x**(1/3)` becomes `nan` ("not a number"). For example, it returns `nan` for $(-8)^{1/3}$ even though the real cube root is -2. Matplotlib skips `nan` points, which erases the left half. The function `np.cbrt` accepts negative inputs and returns -2 for -8. The TikZ version of Figure 1.6 solves the same problem by graphing the left half as $-(-x)^{1/3}$.

Answer to Exercise 8 (on page 14)

It prints `2 0 0`. An even function needs $h(-1) = h(1)$, but $0 \neq 2$. Oddness requires $-h(-1) = h(1)$, which fails for the same reason. A single input that breaks a definition is enough to rule it out.

Answer to Exercise 9 (on page 15)

For `np.abs(x)`, the dashed $h(-x)$ curve lies exactly on the solid curve, and the tests print `True` then `False`: $|x|$ is even. With `x**3`, the dotted $-h(-x)$ curve lies on the solid curve, and the tests print `False` then `True`: x^3 is odd.

Answer to Exercise 10 (on page 17)

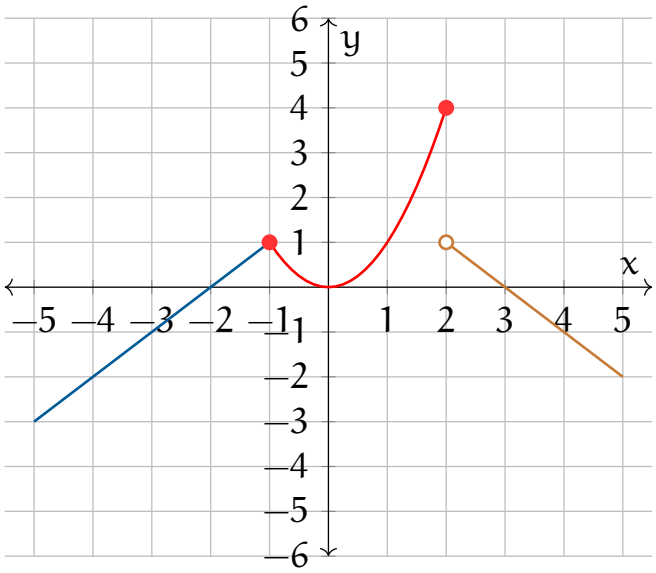
From $x \in [0, \infty)$, we say that $|x| = x$. For $x < 0$, we say $|x| = -x$, since we want to invert the magnitude. So,

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Answer to Exercise 11 (on page 18)

First, let's create a table of values for each integer from -5 to 5:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x)$	-3	-2	-1	0	1	0	1	4	0	-1	-2



Answer to Exercise 12 (on page 20)

1.

$$C(x) = \begin{cases} 400x & 0 < x \leq 5 \\ 2000 + 350(x - 5) & 5 < x \leq 15 \\ 5500 + 300(x - 15) & x > 15 \end{cases}$$

2. $C(12) = 2000 + 350(12 - 5) = \$4450$

3. The average cost is \$360 when $\frac{C(x)}{x} < 360$. This will only happen during the 3rd case, when $x \geq 15$.

$$C(x) = 5500 + 300(x - 15) = 5500 + 300x - 4500 = 1000 + 300x$$

Now,

$$\frac{1000 + 300x}{x} < 360$$

$$300 + \frac{1000}{x} < 360$$

$$\frac{1000}{x} < 60$$

$$1000 < 60x$$

$$x > \frac{1000}{60} \approx 16.67 \rightarrow 17$$

There must be 17 TVs sold for the average cost to fall to \$360

Answer to Exercise 13 (on page 22)

The outputs for $x = -5, -4, \ldots, 5$ are $-3, -2, -1, 0, 1, 0, 1, 4, 0, -1, -2$, matching the table. At $x = -1$ the first condition fails ($-1 < -1$ is false), so Python uses x^2 and returns 1. At $x = 2$ the second condition includes 2, so Python returns $2^2 = 4$. Both are the closed circles on the graph.

Answer to Exercise 14 (on page 22)

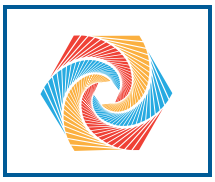
The input 4 satisfies `x >= 0`, so Python returns $4^2 = 16$ and never reaches the `elif`. Python does not flag overlapping cases; it takes the first match. Putting the special case first fixes it:

```
def q(x):
    if x == 4:
        return 10.2
    elif x >= 0:
        return x**2
```

Now `q(4)` returns `10.2` and `q(3)` returns `9`. Ordering the checks this way does the same job as the condition $x \geq 0,\ x \neq 4$ in the revised definition.

Answer to Exercise 15 (on page 23)

It prints `-3 -2 9 7`. Flooring -2.5 moves down to -3, the greatest integer less than or equal to -2.5; rounding toward zero would give -2, which is wrong here. Its ceiling is -2. An input of 8.00001 rises to 9, and an integer such as 7 comes back unchanged.



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